

NAME :

$$f_m = \frac{600}{2\pi} (2\pi \cdot 600) t$$
$$f_c = 1200$$
$$600 \times 2$$

Reg.No: 2403031099

T. No.

1. Which of the following is a disadvantage of SSB modulation?

- A) It consumes more bandwidth than AM. B) It requires complex generation and detection processes
C) It has lower power efficiency. D) It is less reliable in mobile communications

2. An AM wave is described by the equation $s(t) = A_c[1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$. This equation represents which type of AM?

- a) Double Side Band Suppressed Carrier (DSB-SC) b) Single Side Band (SSB)
c) Double Side Band Full Carrier (DSB-FC) d) Vestigial Side Band (VSB)

3. The Phase Modulated signal is given by :

- a) $s(t) = A_c \cos(2\pi f_c t + \beta \cos(2\pi f_m t))$ b) $s(t) = A_m \cos(2\pi f_m t + \beta \cos(2\pi f_c t))$
c) $s(t) = A_c (1 + \mu m(t)) \cos(2\pi f_c t)$ d) $s(t) = A_c \cos(2\pi f_c t) + A_m \cos(2\pi f_m t)$

4. makes the information signal more compatible with the medium

- A. Envelop B. Multiplexing C. Modulation D. Square Law

5. Which devices did we use for AM DSB-full carrier Demodulation

- A) Envelope detector and square law demodulator.
B) Ratio detector and Slope detector
C) coherent
D. Costas level

6. What defines an "ideal filter" according to the provided material?

- a) It allows limited transmission of signals within a broad range.
b) It completely rejects signals within a specific band and passes all others.
c) It has very sharp cutoff characteristics, passing signals exactly within a specified band.
d) It amplifies signals within the passband and attenuates outside.

7. Distortion differs from noise in that distortion:

- a) Is random and unpredictable b) come from imperfect channel response
c) Is caused by external sources d) Only affects digital signals

8. If SSB-SC, the modulating signal has a frequency of 150 Hz, then the bandwidth of the modulated signal is Hz.

- A) 150 B) 75 C) 300 D) 100

9. What are the two primary components of an Envelope Detector, as described in the block diagram?

- a) A Diode and a Band Pass Filter. b) A Diode and a Low Pass Filter.
c) A Summer and a Diode. d) A Local Oscillator and an AM Modulator.

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signals?

A) Bandpass filters

B) Amplifiers

☒ C) Guard bands

D) Demodulators

11. The imaginary line that follows the peaks of the modulated carrier wave and has the shape of the original message signal is called the:

a) Carrier

b) Sideband

c) Envelope

☒ d) Broadband

12. In FM modulation, The ratio of the maximum frequency deviation to that of frequency message, is known as :

A) frequency interval

B) modulation index

☒ C) frequency deviation

D) carrier swing

13. Key advantage of FM over AM is:

a) Smaller bandwidth

b) Immunity to amplitude noise

☒ c) Simpler receivers

d) Longer propagation via ionosphere

14. AM broadcast station transmits modulating frequency up to 6KHz. If transmitting frequency is 810KHz, then maximum and lower sidebands are _____

A. 816KHz and 804KHz

B. 826KHz and 804KHz

☒ C. 916KHz and 904KHz

D. 822KHz and 816KHz

15. In amplitude modulation, frequency and phase of carrier

A. varies simultaneously

B. varies alternately

☒ C. initially varies but become same after time

D. remains constant

☒ 16. What is the primary purpose of a communication system?

a) To amplify signals for long-distance transmission.

b) To convert electrical signals back into messages.

☒ c) To transmit information signals through a communication channel.

d) To reduce noise and interference in a signal.

17. In narrow band FM signal, The maximum frequency deviation is :

A) 75 KHZ

B) 57 KHZ

☒ C) 15 KHZ

D) 51 KHZ

18. Single tone amplitude modulation _____

A. consists of only one frequency component

B. contains a large number of frequency components

☒ C. contains no frequency components

D. contains infinite number of frequency components

19. In FM broadcasting, limited coverage areas result from:

a) Low-frequency operation

b) Line-of-sight propagation

☒ c) Ionospheric reflections

d) Small bandwidth

20. the bandwidth of the VOFSC modulated wave:

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Test 1

Problem :

An audio signal is given by:

$$x(t) = 8 \cos(\pi(600)t)$$

This signal is used to DSB-FC modulation with a carrier signal represented by:

$$c(t) = 40 \cos(\pi(16000)t)$$

Assume the modulation index is 0.1, Determine:

1. Sideband Frequencies.
2. Amplitude of Each Sideband.
3. Bandwidth Required for the AM Signal. $\rightarrow BW = 2f_m$
4. Efficiency of the AM Wave.
5. Draw the Frequency Spectrum of the AM Signal, labeling all components clearly.
6. Total power, carrier power, sideband power

$$x(t) = 8 \cos(\pi(600)t)$$

$$c(t) = 40 \cos(\pi(16000)t)$$

$$\mu = 0.1$$

$$f_m = \frac{600\pi}{2\pi} \rightarrow f_m = 300 \text{ Hz}$$

$$f_c = \frac{16000\pi}{2\pi} \rightarrow f_c = 8000 \text{ Hz}$$

$$\text{①} \rightarrow f_{USB} = f_c + f_m \rightarrow \text{sidebands} = 7700 \text{ Hz}, 8300 \text{ Hz}$$

$$BW = 2f_m \rightarrow \text{Bandwidth} = 600 \text{ Hz}$$

$$A_{SB} = \frac{\mu A_c}{2} \rightarrow \text{Sideband amplitude} = 2$$

$$P_c = \frac{A_c^2}{2R} \rightarrow \text{Carrier power} = 800 \text{ W}$$

$$P_{SB} = \frac{A_{SB}^2}{2R} \rightarrow \text{Sideband power} = 4 \text{ W}$$

$$P_t = P_c + P_{SB} \rightarrow \text{Total power} = 804 \text{ W}$$

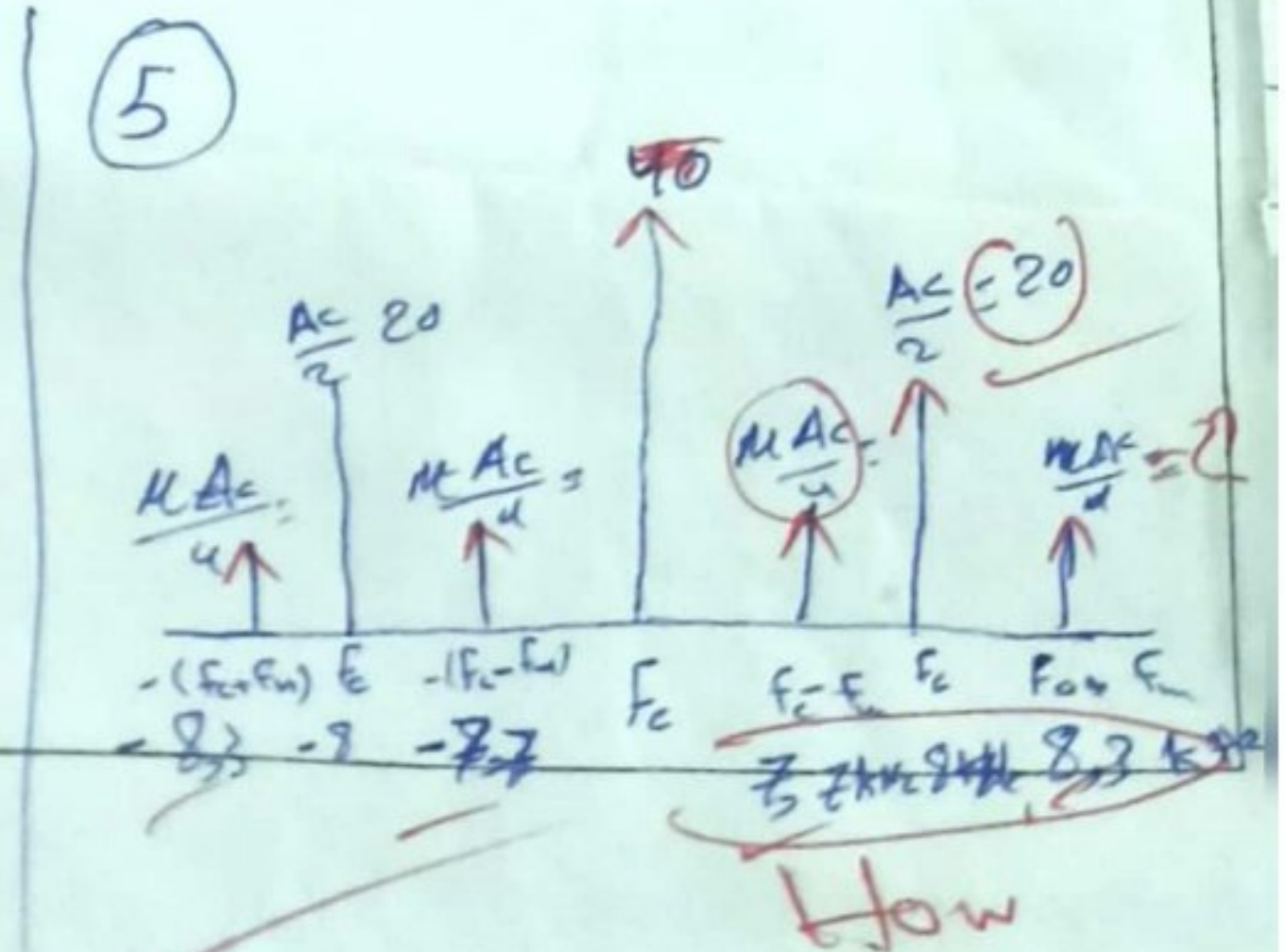
$$\eta = \frac{P_{SB}}{P_t} \rightarrow \text{Efficiency} = 0.5\%$$

$$\text{⑥} \rightarrow P_c = \frac{A_c^2}{2R} = \frac{(40)^2}{2(1)} = 800 \text{ W}$$

$$P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$$

$$800 \left(1 + \frac{(0.1)^2}{2}\right) = 804 \text{ W}$$

$$P_{SB} = P_c \frac{\mu^2}{2} = 4 \text{ W}$$



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